

LoRA/QLoRA for STEM Solution Generation: Evidence Review for Project "Aryabhata"

TL;DR

- LoRA/QLoRA can work for STEM solution generation — but as *distillation/elicitation* of capability the base model already has, not as knowledge injection. Your project's exact namesake proves it: Physics Wallah's Aryabhata 1.0 used LoRA (rank 128) on a merged 7B math model and hit 86.0%/90.2% on JEE Main Maths (Jan/Apr 2025), beating GPT-4o (which scored just 43.55% on the April papers). With a few hundred to ~1,000 examples per subject you can meaningfully shift behavior — but only if (a) the base model already "knows" the math, (b) you train on *verified* chain-of-thought (CoT) solutions, and (c) you curate ruthlessly.
- LoRA is strong for format/style/reasoning-elicitation and weak for injecting genuinely new factual knowledge. "LoRA Learns Less and Forgets Less," "Fine-Tuning or Retrieval?", and Arcee.ai's CPT study all show LoRA/PEFT underperforms full fine-tuning and RAG when the task requires absorbing facts the model never saw. But Thinking Machines' "LoRA Without Regret" shows LoRA *matches* full fine-tuning on small/medium post-training datasets when applied to **all layers (including MLP)** at a learning rate ~10× higher than full fine-tuning.
- For your Phase 1 (distilling Gemini 3.1 Pro quality into Gemini 2.5 Flash), the dominant lever is data quality, not the PEFT method. The strongest small-data evidence (s1: 1,000 examples; LIMO: 817 examples) and the strongest distillation evidence (DeepSeek-R1; Phi-4-mini-reasoning) all converge: a few hundred to ~1K carefully curated, answer-verified CoT traces from a strong teacher can elicit competition-level reasoning — *provided the student base is already math-capable*.

Key Findings

1. Your project's exact analogue already exists and used LoRA. Physics Wallah's Aryabhata 1.0 (7B, JEE Main Math) used LoRA rank 128 / alpha 128 / dropout 0.1 on all attention + MLP modules for its SFT stage — explicitly *not* full fine-tuning ("We used Parameter Efficient Finetuning, particularly Low-Rank Adaptation... during SFT"). It scored 86.0% (Jan 2025) and 90.2% (Apr 2025) on JEE Main Maths and 94.8% GSM8K / 83.6% MATH-500. On the April JEE papers its 90.2% beat GPT-4o (43.55%) and GPT-4.1 (80.44%). (Storyboard18)

2. **Small data CAN elicit strong reasoning — if the base model is already strong.** s1 (1,000 examples distilled from Gemini Thinking → Qwen2.5-32B) and LIMO (817 examples → Qwen2.5-32B) both prove ~1K examples can *elicit* latent reasoning. This is the "less-is-more / superficial alignment" hypothesis: pre-training installs the capability; fine-tuning activates the format.
3. **Distillation from a stronger teacher is the proven recipe for Phase 1.** DeepSeek-R1 distillation (800K teacher traces → Qwen/Llama) lifted small models dramatically; the Phi-4-mini-reasoning paper notes R1-style distillation lifted Llama-8B MATH-500 from 44.4%→89.1% via SFT on synthetic CoT alone.
4. **LoRA is the wrong tool for genuine knowledge injection.** Ovadia et al. found RAG more than doubled new-fact accuracy vs fine-tuning; Arcee.ai found full continued-pretraining beat QLoRA CPT for new domain knowledge.
5. **Rank and target-module choice are decisive for reasoning.** Apply LoRA to ALL layers (especially MLP), use LR ~10× full-FT. Rank 8 can fail to learn reasoning where rank 256 succeeds.

Details

POSITIVE CASES (weighted toward LLM/STEM reasoning)

Case 1 — Aryabhata 1.0 (Physics Wallah) — the single most relevant case.

- *Need / domain:* JEE Main Mathematics step-by-step solutions for Indian competitive-exam prep — essentially your Phase-1/Phase-2 target.
- *Base model:* A linear model-merge ("model soup," via MergeKit) called Aryabhata 0.5, combining **Qwen2.5-Math-7B-Instruct + AceMath-7B-Instruct (NVIDIA's Qwen-Math fine-tune) + DeepSeek-R1-Distill-Qwen-7B** (the HF card lists base_model Qwen/Qwen2.5-Math-7B). Merge weights $\alpha + \beta + \gamma = 1$ chosen empirically; exact coefficients not published. (Hugging Face) (arXiv)
- *Method:* **LoRA / PEFT, rank 128, alpha 128, dropout 0.1**, target modules q/k/v/o + gate/up/down + embeddings. SFT hyperparameters: bf16, max sequence length 16,384, per-device batch $1 \times$ grad-accum 16 (effective batch 16), 3 epochs, LR $2e-5 \rightarrow 2e-7$ linear, AdamW-8bit, 5 warmup steps. Followed by **RLVR** (custom GRPO with KL penalty and clipping removed; 4,096-token context). (huggingface)
- *Data curation (concrete numbers):* ~250K raw questions extracted → filtered (removed diagram-based, non-English, option-heavy) → ~130K curated → **best-of-4 rejection sampling** with Aryabhata 0.5, keeping only CoTs that reach the correct answer → **~350K verified CoTs across ~100K questions** used for SFT (curriculum-ordered easy→hard; the 67,247 questions with 4/4 correct CoTs were 10%-sampled). The 31,470 questions with 0/4 correct CoTs were reserved for RLVR. (Hugging Face)

- **Results:** JEE Main Maths **86.0% (Jan 2025), 90.2% (Apr 2025)**; on April papers beat GPT-4o (43.55%) and GPT-4.1 (80.44%). Out-of-distribution: **MATH-500 83.6, GSM8K 94.8** (vs GPT-4o 69.2/94.6; Gemini 2.5 Flash 93.6/85.1; its base Qwen2.5-Math-7B-Instruct 66.0/94.7). Token-efficient at ~2K tokens/response vs the long traces of o1/R1-style models. Trained on a single 1×2 H100 setup. *Takeaway: LoRA-based SFT + RLVR on a merged math base, with rigorous rejection-sampled CoT, is a directly transplantable blueprint.* (Analytics India Magazine + 2)

Case 2 — s1 / s1K (Stanford, Muennighoff et al., arXiv 2501.19393). 1,000 questions + reasoning traces distilled from Gemini Thinking → SFT of Qwen2.5-32B-Instruct (full FT, 26 min on 16 H100s). Verbatim: "our model s1-32B exceeds o1-preview on competition math questions by up to 27% (MATH and AIME24)." With "budget forcing," AIME24 rose from 50%→57%. Proves ~1K curated examples + strong base elicits competition reasoning.

(OpenReview + 2)

Case 3 — LIMO (arXiv 2502.03387). 817 curated examples → Qwen2.5-32B-Instruct (full FT, 15 epochs, LR 5e-6 cosine). AIME24 57.1% (from 6.5% baseline SFT), MATH 94.8%, average 72.1% — beating o1-preview (67.8%) using ~1% of typical data. Explicit "Less-Is-More Reasoning Hypothesis": when domain knowledge is already in pre-training, minimal demonstrations elicit it.

Case 4 — DeepSeek-R1 distillation (arXiv 2501.12948). 800K R1-generated traces → SFT (no RL) of Qwen2.5-Math/Llama bases. R1-Distill-Qwen-1.5B: 83.9% MATH-500, 28.9% AIME pass@1 (52.7% cons@64); 7B: 92.8% MATH-500, 55.5% AIME, beating GPT-4o/Claude-3.5 on math. Pure distillation SFT, no LoRA.

Case 5 — Phi-4-mini-reasoning (Microsoft, 3.8B, arXiv 2504.21233). Multi-stage distillation on DeepSeek-R1 synthetic CoT (batch 128, LR 1e-5, 5 epochs, 16K→20K seq len) + RL. MATH-500 94.6, AIME-2024 57.5 — beating R1-Distill-Qwen-7B (91.4/49.5) at half the size. Phi-4-reasoning (14B) used o3-mini as teacher and scored 75.3% AIME-2024, beating R1-Distill-70B. Demonstrates "data-first" curated-CoT distillation. (arXiv + 3)

Case 6 — Qwen2.5-Math (arXiv 2409.12122). The leading open math base family and Aryabhata's foundation. Self-improvement pipeline: ~1T-token math corpus, rejection-sampling SFT (1M+ CoT problems), reward model, RL. 7B base scores 91.6 GSM8K / 55.4 MATH; instruct 7B hits 83.6 MATH (CoT). *Pick this family as your Phase-2 base.*

Case 7 — LoRA Land (Predibase, arXiv 2405.00732). 310 QLoRA (4-bit) models across 10 bases and 31 tasks; fine-tuned models beat base by **+34 points** and GPT-4 by **+10 points** on average (+38.7 avg lift from fine-tuning). *Caveat:* math (GSM8K) was among the weakest tasks — a published Mistral-7B GSM8K LoRA scored only ~0.37 on a math eval; GPT-4 still

won 6/31 tasks, all "broader, more complex domains." LoRA shines on narrow/structured tasks, struggles on open math. (arXiv)

Case 8 — "LoRA Without Regret" (Thinking Machines, Schulman et al., Sep 2025).

Systematic Llama-3/Qwen-3 experiments: for small-to-medium SFT and reasoning datasets, **LoRA matches full fine-tuning** if (1) applied to all layers, especially MLP/MoE (attention-only underperforms even at matched parameters), and (2) LR is set $\sim 10\times$ full-FT ("an optimal multiplier around $15\times$ for short runs, converging to $\sim 10\times$ for longer runs"). For RL: "LoRA performs equivalently to FullFT... even with small ranks... RL requires very low capacity" — rank 1 can match. LoRA underperforms only at pretraining-scale datasets that exceed adapter capacity. (Thinkingmachines) (thinkingmachines)

Case 9 — Image/diffusion LoRA (context for the "style vs capability" distinction). Stable Diffusion / Flux LoRA practice: character LoRAs 40–80 images, style LoRAs 25–50, concept LoRAs 30–60; typical rank ~ 32 , alpha 16, LR $1e-4$, 2,000–5,000 steps. The Calvin-and-Hobbes style-transfer paper and building-façade LoRA (200 images, 1 epoch) confirm LoRA's sweet spot is **style/concept adaptation with tiny data** — the diffusion analogue of why LoRA is great for *format* and weak for *new knowledge*.

NEGATIVE / CAUTIONARY CASES

Neg 1 — "LoRA Learns Less and Forgets Less" (Biderman et al., TMLR 2024, arXiv 2405.09673). On programming and math at $\sim 100K$ -example IFT and $\sim 20B$ -token continued-pretraining (CPT) regimes, LoRA **substantially underperformed full FT** and was less sample-efficient. In CPT the gap never closed even at high rank (best LoRA $r=256$ peaked at GSM8K 0.203 vs full-FT 0.293). In *math IFT* the gap was smaller — $r=64$ reached GSM8K 0.624 vs full-FT 0.642 — because math IFT is a smaller domain shift from pre-training. LoRA forgets less (regularization benefit). Best practices: tune LR ($\sim 10\times$ higher), target MLP, choose rank carefully; full FT finds weight changes 10 – $100\times$ higher rank than typical LoRA. (arXiv)

Neg 2 — "Fine-Tuning or Retrieval?" (Ovadia et al., EMNLP 2024, arXiv 2312.05934).

Unsupervised fine-tuning failed to inject *new facts*; RAG more than doubled scores on new knowledge (e.g., 0.875 vs 0.504 for Mistral on a current-events task). Fine-tuning only worked when each fact appeared in many paraphrases. *Relevance: don't expect LoRA to teach Flash facts/formulas it doesn't already encode — give it the reasoning format instead.*

Neg 3 — Arcee.ai QLoRA CPT study. On a 2.5B-token SEC dataset, standard full CPT beat QLoRA-based CPT on perplexity. Conclusion: QLoRA is "incapable of introducing the significant new knowledge required in CPT"; it excels at instruction tuning and DPO (which require only minor parameter changes), not knowledge injection. (Arcee) (Arcee)

Neg 4 — "Plasticity vs Rigidity" (Khan et al., EACL 2026, arXiv 2601.06677). Micro-budget RLVR + LoRA on $\leq 1.5\text{B}$ models (single A40, $< 24\text{h}$): **rank-8 LoRA consistently failed** to capture reasoning dynamics; **rank-256 unlocked plasticity** in generalist instruct models (best: AIME24 40.0% pass@1, +11.1 pts). But heavily math-specialized models (Qwen2.5-Math-1.5B) suffered *performance collapse* — noisy low-budget RL acted as destructive interference. *Lesson: for reasoning, use high rank; and don't aggressively RL a model already near its task optimum.*

Neg 5 — Anyscale Llama-2 study. The largest full-FT-vs-LoRA gap was on GSM8K math — "a challenging new skill... not best captured by a low-rank approximation." LoRA matched full FT only at 70B (where the base is already strong), and even then absolute gains were small. Also flagged LoRA's sensitivity to LR for stable training.

Neg 6 — Naive small-model GSM8K LoRA (HF forum). GPT-Neo 2.7B + LoRA on GSM8K produced plausible reasoning steps but $\sim 2\text{--}3\%$ final-answer accuracy. *A too-weak base cannot be rescued by LoRA — the capability has to be latent already.*

Specific evidence on your implicit questions

- **Can a few hundred-1K examples meaningfully improve real problem-solving (not style)?** Yes — if the base already encodes the math (s1: 1K; LIMO: 817; Aryabhata's curriculum SFT). No — if you're trying to teach math the base can't do (GPT-Neo on GSM8K).
- **Style/format vs new capability?** LoRA strongly favors style/format/reasoning-*elicitation*; knowledge injection is its documented weakness (Ovadia, Arcee, Biderman-CPT).
- **Dataset sizes typically needed?** Reasoning *elicitation* from a strong base: $\sim 800\text{--}1,000$ high-quality verified traces. *Distillation to frontier levels*: tens of thousands to 800K traces (R1: 800K; Aryabhata: $\sim 350\text{K}$ CoTs; Phi-4: 1M+ CoT problems). There is no clear saturation point for high-quality math SFT data — more curated data keeps helping.
- **Best practices for math/reasoning LoRA:** train on verified CoT (rejection sampling / best-of-N to keep only correct solutions); rank 64-256 for reasoning ($\alpha \approx \text{rank}$); LoRA on ALL layers including MLP; LR $\sim 10\times$ your full-FT LR; curriculum ordering (easy $\rightarrow$ hard); answer-checking evaluation (float/string match + LLM-as-judge), not just loss. For RL, rank can be very low.
- **Distillation (your Phase 1):** Strongly validated — R1, Phi-4, s1, and Aryabhata all distill from a stronger teacher into a smaller student. Distilling Gemini 3.1 Pro CoT into Gemini 2.5 Flash is the same pattern.
- **Qwen / DeepSeek / Gemma for math:** Qwen2.5-Math is the strongest open base; DeepSeek-R1-Distill-Qwen variants are top small reasoners; Gemma supports LoRA

(Google's own Keras tutorial starts at rank 4 and scales up) but lags Qwen-Math on competition math.

- **Indian-exam AI specifically:** Aryabhata 1.0 is the canonical case; Aryabhata 2.0 (adding Physics and Chemistry, and JEE Advanced/NEET) is planned under a partnership with Microsoft Research India.

What Worked / Why It Failed — Synthesis Table

Case	Method / data size	Base	Outcome	Why it worked / failed
Aryabhata 1.0	LoRA r128 + RLVR; ~350K verified CoT (~100K Qs)	Qwen2.5-Math+AceMath+R1-Distill merge	86-90% JEE; 94.8 GSM8K / 83.6 MATH	✅ Strong math base + rejection-sampled correct CoT + curriculum + RLVR
s1	Full FT; 1,000 examples	Qwen2.5-32B-Instruct	+27% vs o1-preview on math	✅ Capability latent in base; elite curation (difficulty/diversity/quality)
LIMO	Full FT; 817 examples	Qwen2.5-32B-Instruct	AIME 6.5→57.1%	✅ Eliciting, not teaching; curated "cognitive templates"
R1 / Phi-4 distill	SFT; 800K / 1M+ CoT	Qwen-Math, Llama, Phi	Frontier-level small models	✅ High-quality teacher CoT at scale
LoRA Land	QLoRA 4-bit; per-task	10 bases ≤7B	+34 vs base, but math weakest	⚠️ Great on narrow/structured tasks; open math is hard
LoRA Without Regret	LoRA all-layers, 10× LR	Llama-3/Qwen-3	= full FT on small/med data	✅ Right config (MLP layers, LR, rank)
LoRA Learns Less	LoRA vs full FT; 100K/20B	Llama-2	LoRA underperforms, esp. CPT	❌ Pretraining-scale knowledge exceeds adapter capacity
Fine-Tuning or Retrieval	Unsup. FT vs RAG	Mistral/Orca	RAG 2× better on new facts	❌ Fine-tuning poor at new-fact injection

Case	Method / data size	Base	Outcome	Why it worked / failed
Arcee QLoRA CPT	QLoRA vs full CPT; 2.5B tok	—	Full CPT wins	✗ QLoRA can't add major new knowledge
Plasticity vs Rigidity	LoRA+RLVR; r8 vs r256	≤1.5B	r8 fails, r256 works; math-specialists collapse	⚠ Rank too low fails; over-specialized base collapses
GPT-Neo GSM8K	LoRA	GPT-Neo 2.7B	~2-3% accuracy	✗ Base too weak; capability not latent

Decision rule for you: LoRA/QLoRA is the *right* approach for STEM solution generation **when** (1) you start from a math-strong base, (2) you train on *verified-correct* CoT solutions (distilled from a strong teacher), (3) you use sufficient rank (64–256) on all layers with a tuned LR, and (4) you're *eliciting/formatting* reasoning the base can already do. It is the *wrong* approach if you expect it to inject new facts/formulas a weak base lacks — for that, use a stronger base, full fine-tuning/continued pretraining, or RAG.

Recommendations (staged, with thresholds)

Stage 0 — De-risk with a pilot (now): Distill ~1,000 verified CoT solutions per subject from Gemini 3.1 Pro (or another strong teacher), keeping only answer-correct traces. SFT Gemini 2.5 Flash (or, for Phase 2, Qwen2.5-Math-7B). *Threshold to proceed:* pilot must close ≥50% of the Flash↔Pro accuracy gap on a held-out JEE/CBSE set. This mirrors s1/LIMO.

Stage 1 — Scale data, not just method: If the pilot helps but plateaus, expand to ~30K–100K+ verified CoTs per subject via rejection sampling / best-of-N (Aryabhata used ~350K CoTs total). More curated data keeps helping math accuracy with no clear saturation. Order curriculum easy→hard.

Stage 2 — Configure LoRA correctly (Phase 2 / open models): rank 64–256, alpha≈rank, dropout ~0.05–0.1, target ALL layers including MLP (q/k/v/o + gate/up/down), LR ~10× your full-FT LR (or copy Aryabhata: r128/alpha128, LR 2e-5, 3 epochs, seq len 16K, AdamW-8bit). Use QLoRA (4-bit) only for memory limits — it's fine for SFT but avoid it for any continued-pretraining/new-knowledge step.

Stage 3 — Add RLVR after SFT: On the hard questions where SFT CoTs fail (Aryabhata reserved 31,470 such questions), apply GRPO-style RL with verifiable (answer-checking)

rewards. RL needs very low LoRA rank. *Do not* aggressively RL an already math-specialized base (collapse risk).

Stage 4 — Evaluate by correctness, not loss: Use float + string match plus LLM-as-judge for equivalence (Aryabhata's setup). Report pass@1 at temperature 0.

Benchmarks that should change your plan:

- If LoRA SFT plateaus materially below full fine-tuning on held-out JEE/MATH → switch to full fine-tuning (or raise rank toward 256 and verify MLP layers are targeted).
- If accuracy stays low even with full FT → your base is too weak; switch to a math-stronger base (Qwen2.5-Math / R1-Distill / merge).
- If the model "knows the steps but gets the number wrong" persistently → add tool-integrated reasoning (Python execution) à la Qwen2.5-Math TIR, rather than more SFT.

Caveats

- **Method confound in the headline small-data cases:** s1 and LIMO used *full* fine-tuning on 32B models, not LoRA — their *data-efficiency* lesson transfers to your setting, but the LoRA-specific method does not 1:1. Aryabhata is the one that combined small-data philosophy *with* LoRA.
- **Aryabhata's published detail is partial:** the merge coefficients (α , β , γ) are not disclosed, and the per-competitor JEE accuracies live mostly in a figure; the GPT-4o 43.55% / GPT-4.1 80.44% April-session numbers come via Storyboard18's reporting of PW's claims (arXiv 2508.08665). (Storyboard18)
- **Phase-1 platform constraint:** Gemini 2.5 Flash is a closed API model; you likely cannot run open-source-style LoRA on it directly. Google's managed tuning may behave differently from the open LoRA literature, and "Gemini 3.1 Pro" was not independently verifiable as of this writing — treat the teacher↔student gap empirically.
- **Generalization:** strong in-distribution JEE/MATH gains don't guarantee out-of-distribution robustness; keep a forgetting check on general-language benchmarks (LoRA's known advantage is that it forgets *less* than full FT).
- **Recency/uncertainty:** several cited works are very recent (2025–2026 preprints/workshop papers); the "rank 8 fails / 256 works" and "13-parameter TinyLoRA" findings are intriguing but from small studies — treat as directional, not settled.

